

7(a)	1 calculate R_f value(s) ; 2 $(R_f =) \frac{\text{distance moved by pigment (from baseline)}}{\text{distance moved by solvent (from baseline)}} ;$ 3 compare with, known / standard / table, values ;	3
7(b)	any three from: 1 absorb light wavelengths not absorbed by, reaction centre / primary pigment / chlorophyll a ; A harvest 2 so extend the range of wavelengths absorbed ; 3 pass energy to, reaction centre / primary pigment / chlorophyll a ; 4 <i>ref. to</i> accessory pigments ;	3

Question	Answer	Marks
7(c)	any seven from: 1 carbon dioxide, reacts / combines, with, ribulose biphosphate / RuBP ; 2 (catalysed by) rubisco ; 3 <i>ref. to</i> carbon (dioxide) fixation ; 4 unstable 6C compound ; 5 forms 2 (molecules of) glycerate 3-phosphate / GP ; 6 glycerate 3-phosphate / GP, reduced to, triose phosphate / TP ; 7 by reduced NADP and ATP ; 8 triose phosphate / TP, used to regenerate RuBP ; 9 glycerate 3-phosphate / GP, forms amino acids ; 10 (some) TP forms, (named) hexoses / sucrose / maltose / starch / cellulose / glycerol / lipids / amino acids ; 11 <i>ref. to</i> Calvin cycle ;	7